

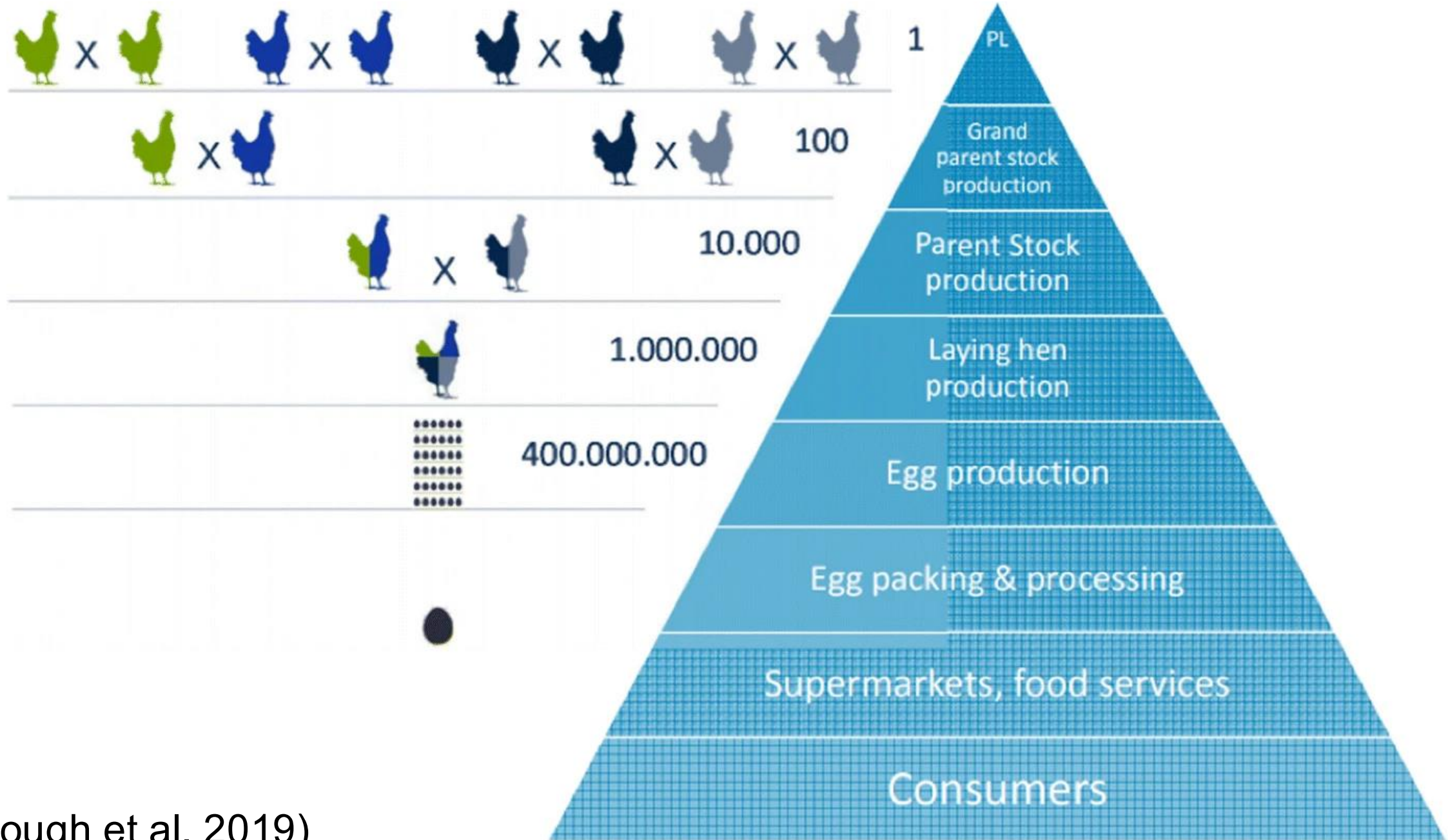
Simulation of breeding programs: chicken

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Agenda

1. A few words about poultry breeding
2. Back to basics: Dadousis et al. (2021)
3. Complex breeding structures: de Hollander et al. (2024)
4. Evaluation and genomics

A bit of poultry breeding



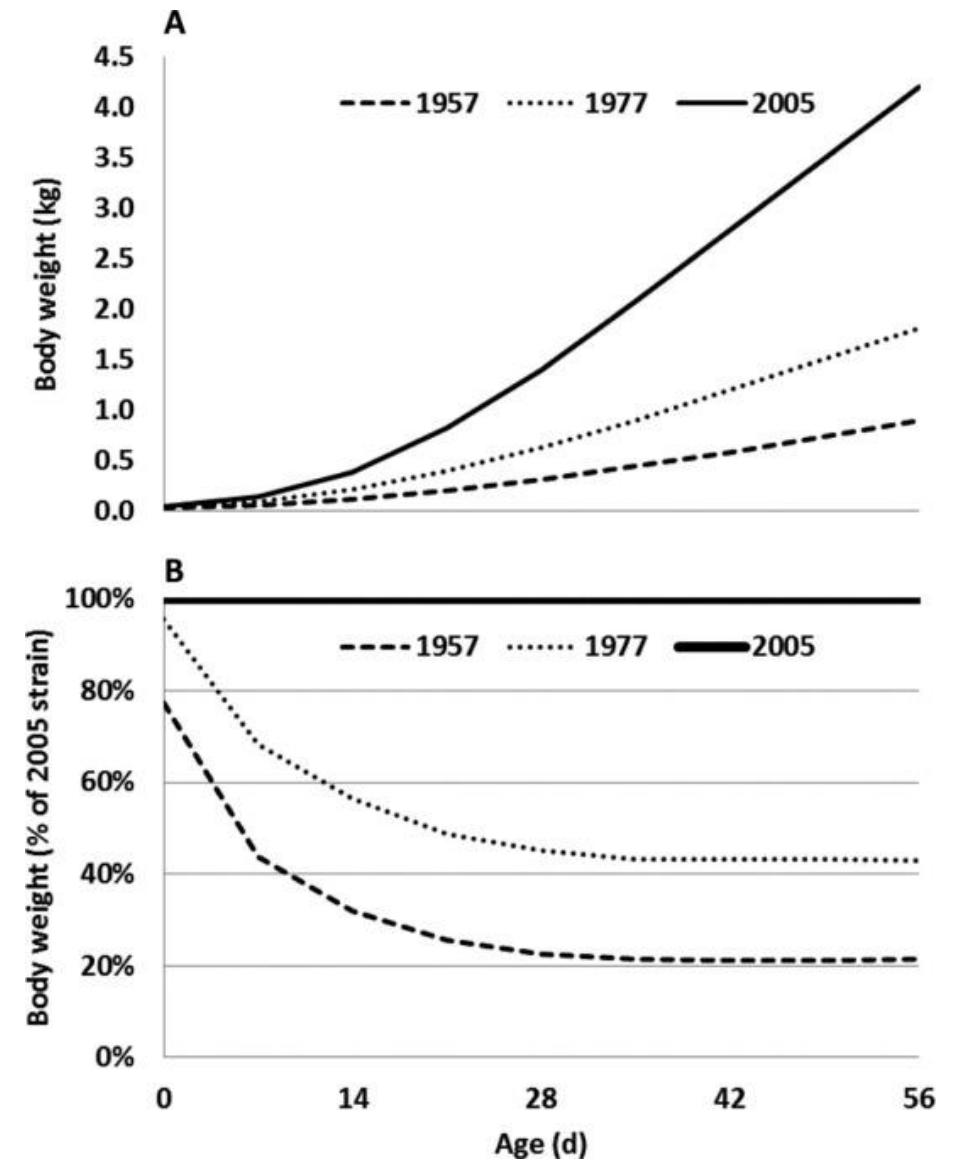
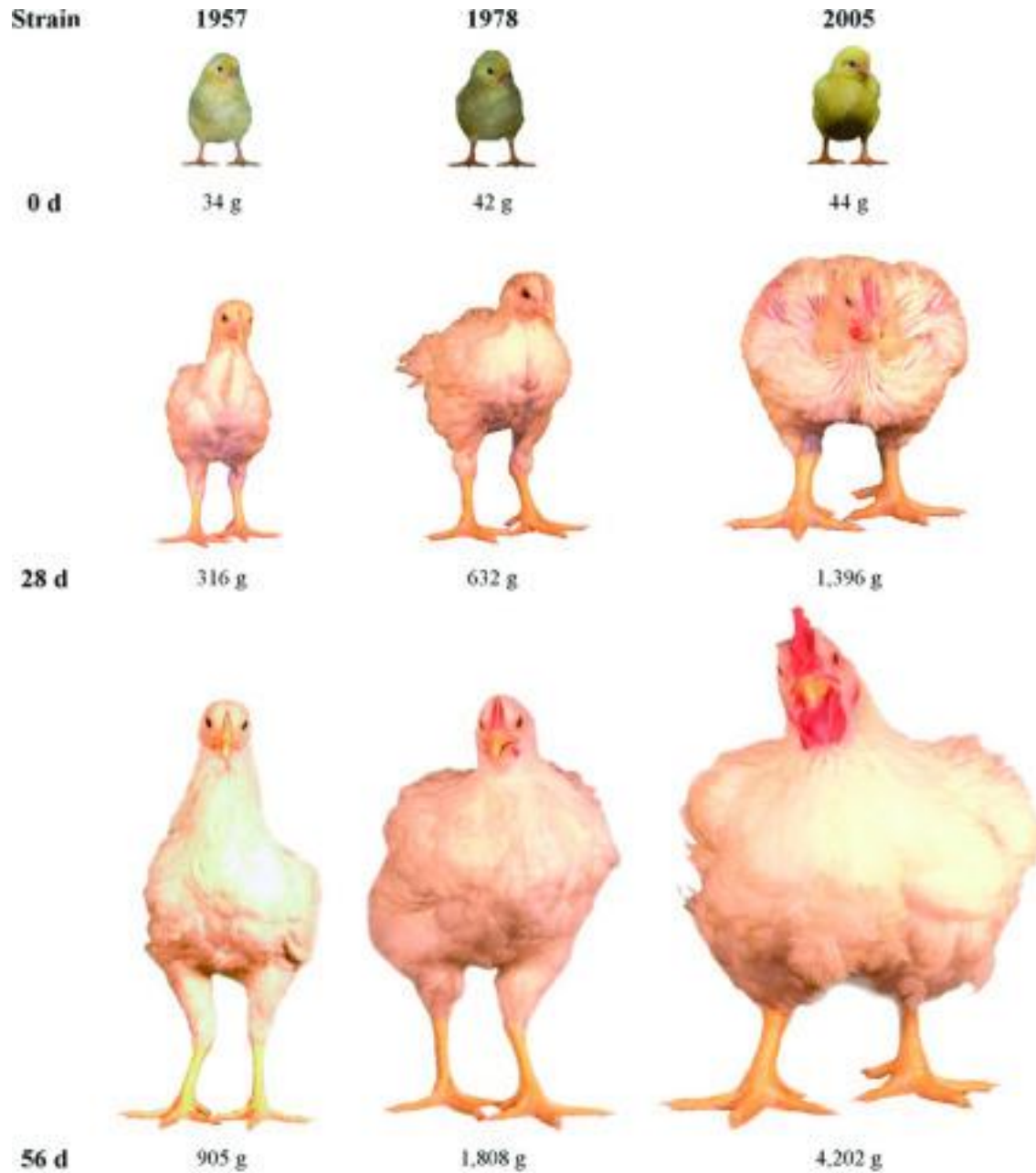
(FERNYHOUGH et al. 2019)

Modern poultry breeding started in North America

Hybrid strategies for chickens and laying hens 1940-1960

Centralised, with few international breeding companies

(Leenstra et al. 2016;
Derry 2012)



(Zuidhof et al. 2014)

Breeding happens in purebred lines

Effectively small (but bigger in broiler than layer)

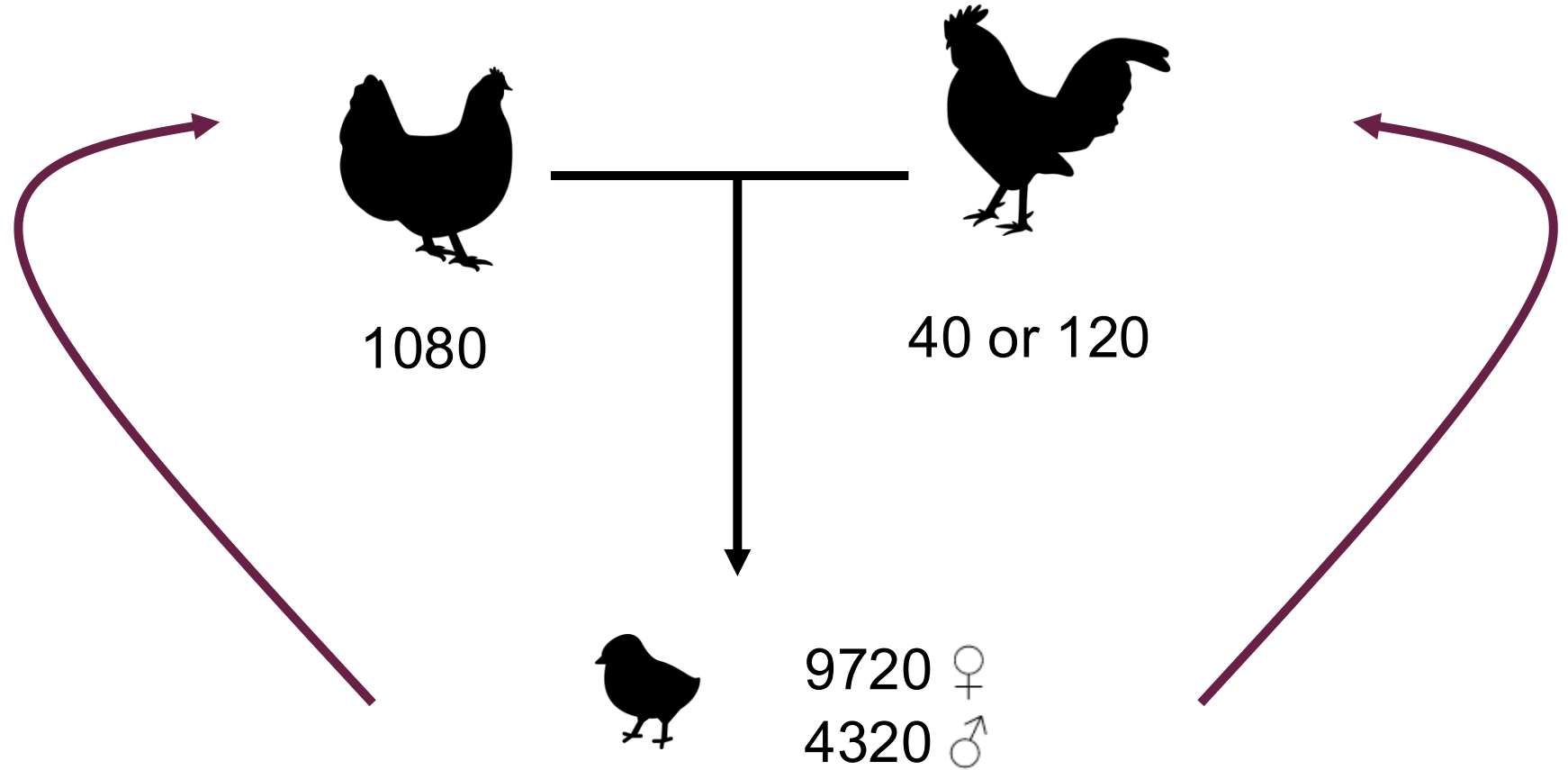
Breeding companies keep many lines

Selection on some combination of:

- Own phenotype from purebreds
- Information from their relatives
- Crossbred information – topic of research
- Individual or family groups

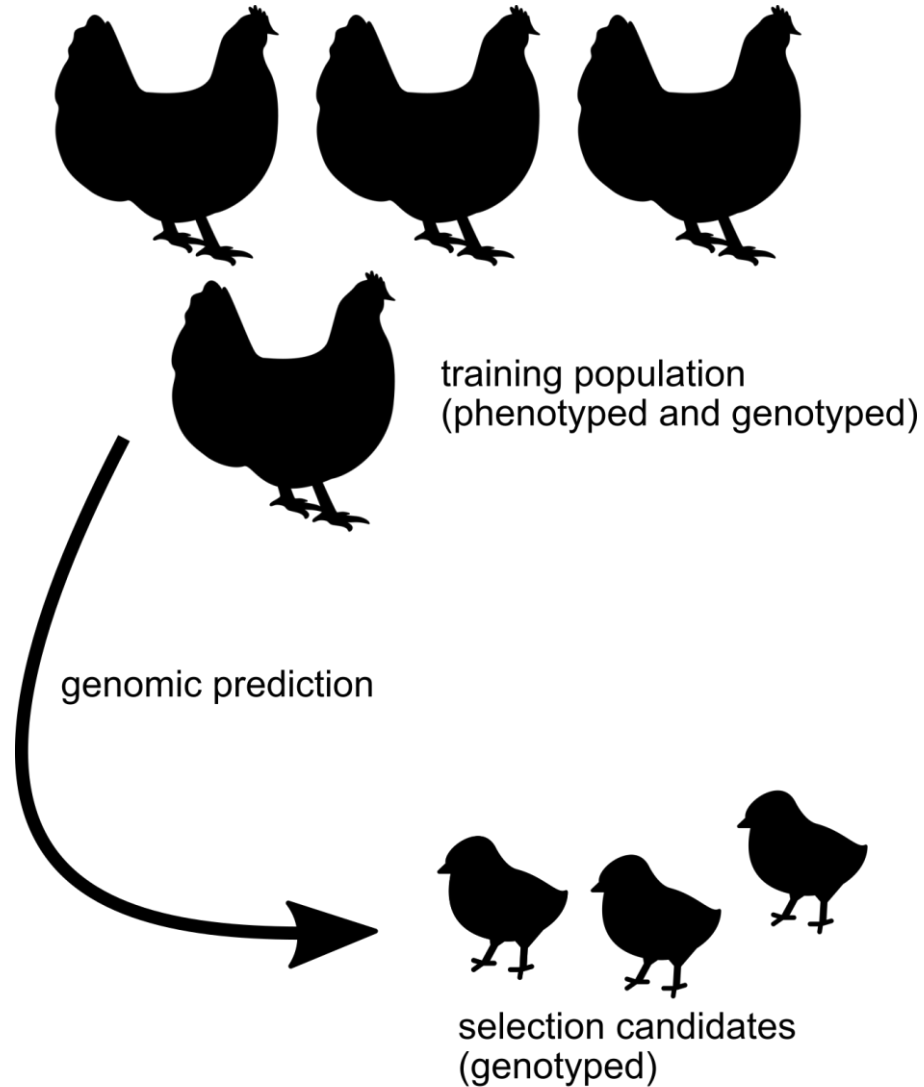
Simulated breeding program from one line

Without genomic prediction



One round of selection per year

(Pocrnic et al. 2023)



(Images: Freepik from flaticon.com)

Genomic prediction

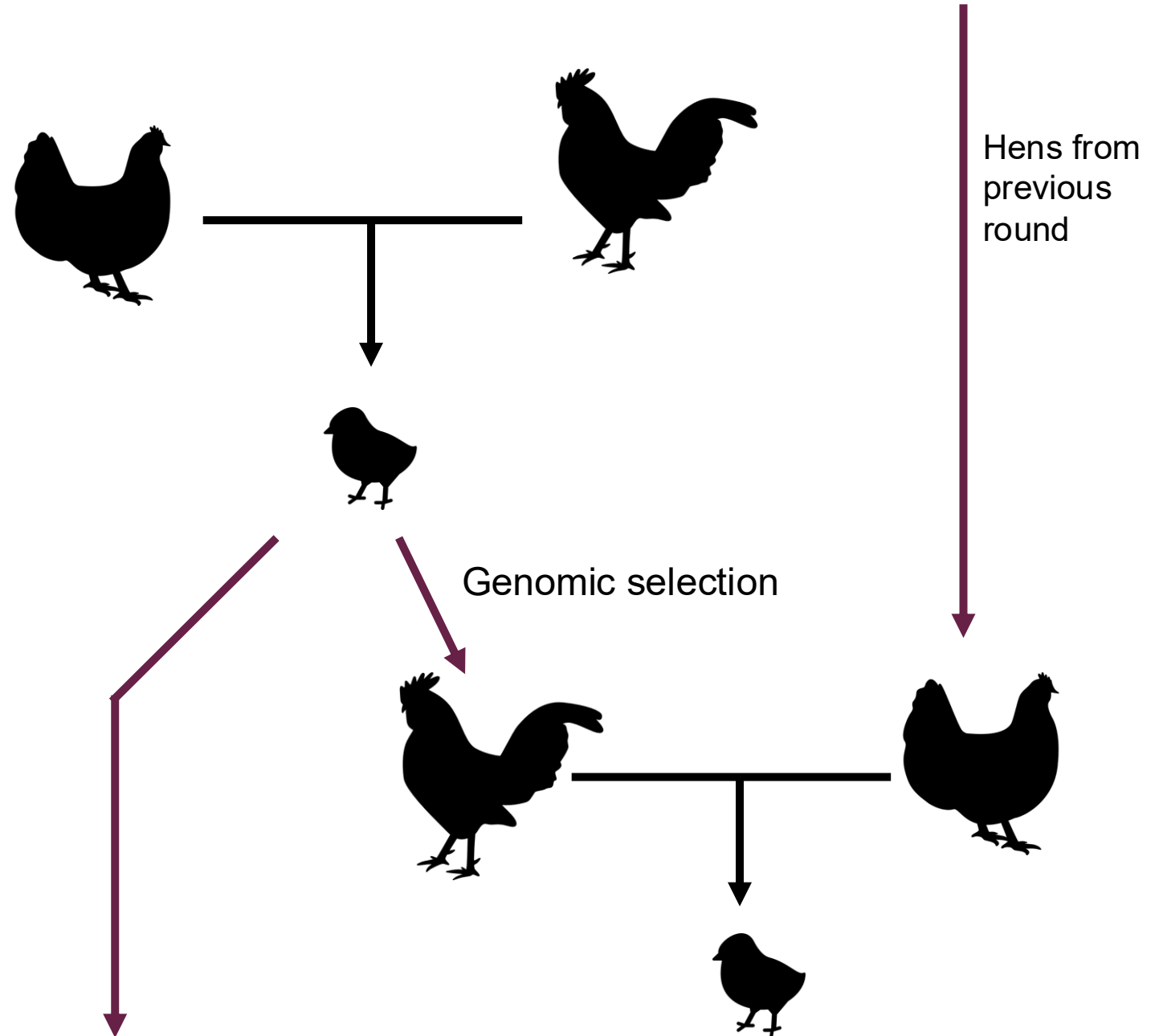
Genetic evaluation using DNA information to propagate information from relatives, in addition to pedigree

Layers: increase accuracy and generation time (somewhat)

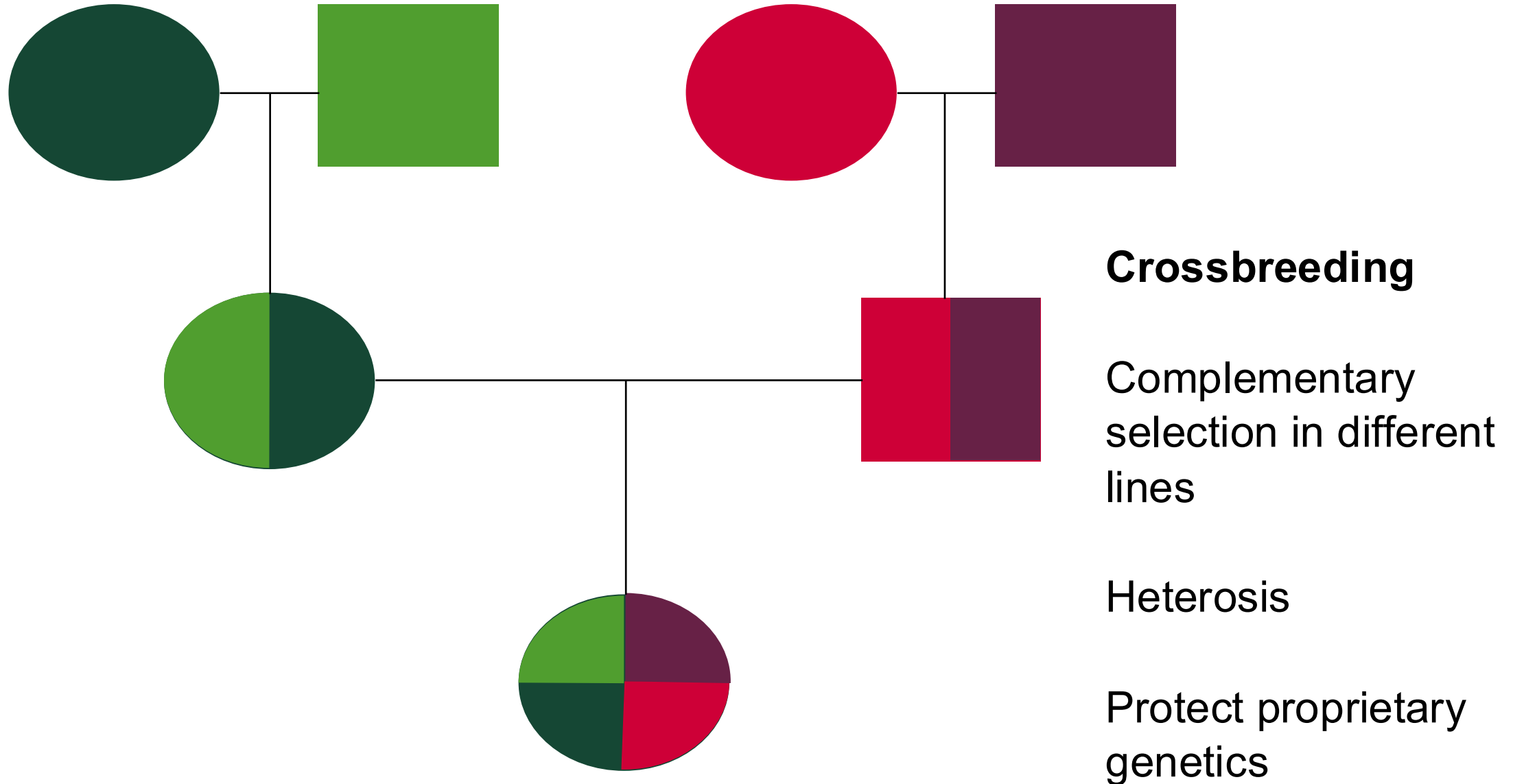
Broilers: increase accuracy but not generation time

(Wolc et al. 2016)

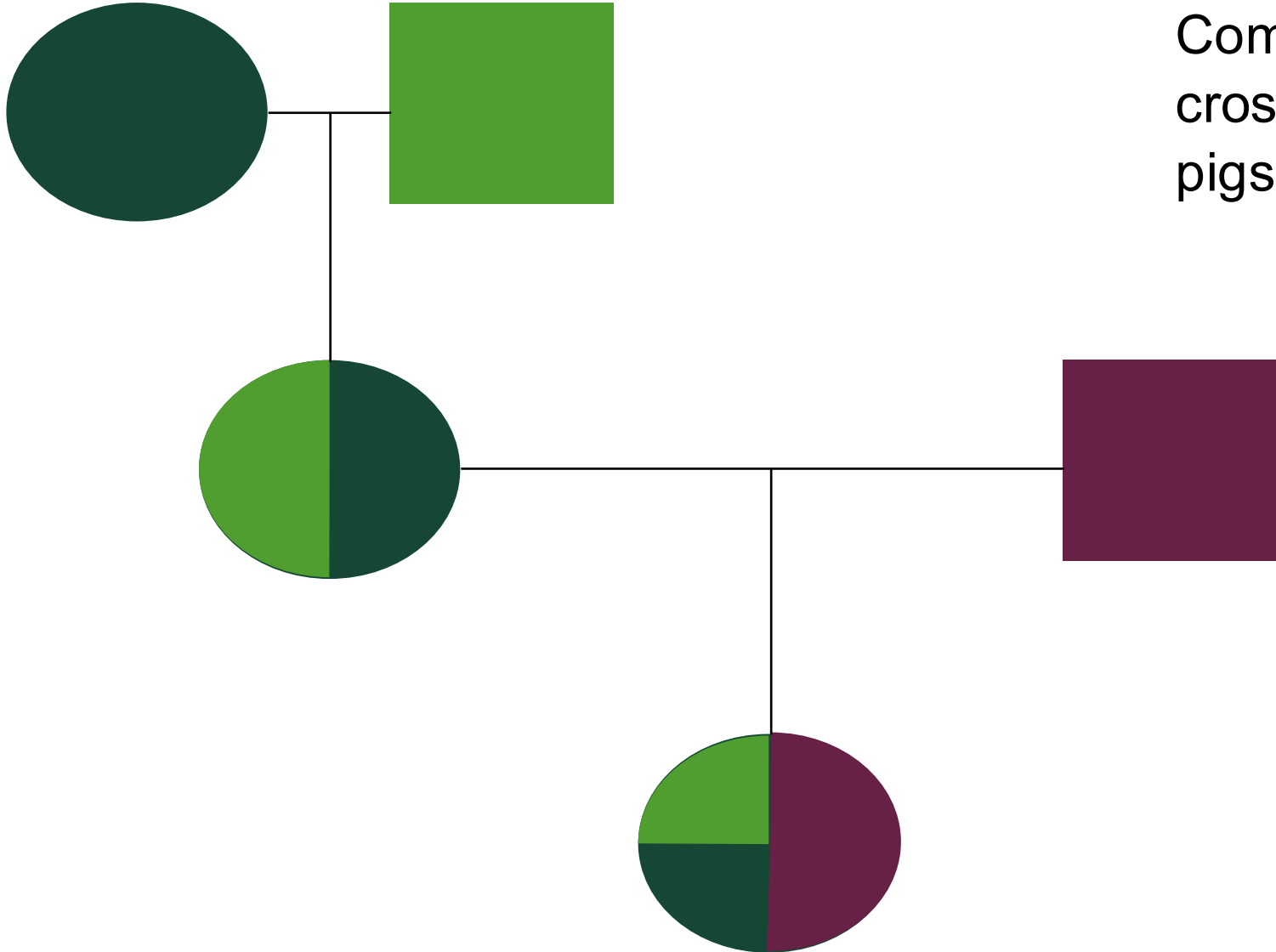
Genomic prediction allowing
two rounds of selection (on
male side)



(Pocrnic et al. 2023)



Compare to 3-way
crossbreeding used in
pigs



Back to basics: Dadousis et al. (2021)

Goal:

- Illustrate quantitative genetic concepts
- Parameters from Dadousis et al. (2021), GWAS of body weight in a broiler chicken breeding program

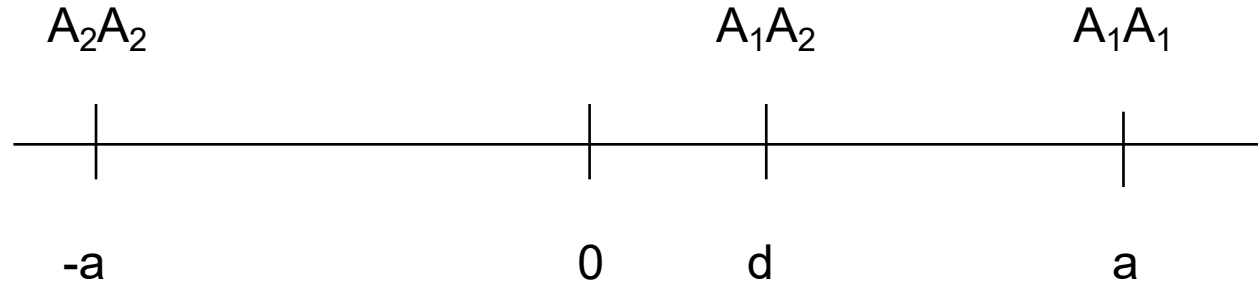
The quantitative genetic standard model

$$P = G + E$$

$$P = A + D + E + I$$

$$P \approx A + E$$

Locus-level coefficients (Falconer & Mackay 1996)



$$A_1A_1: A_{A_1A_1} = \alpha_{A_1} + \alpha_{A_1} = 2q\alpha$$

$$A_1A_1: D_{A_1A_1} = -2q^2d$$

$$A_1A_2: A_{A_1A_2} = \alpha_{A_1} + \alpha_{A_2} = (q - p)\alpha$$

$$A_1A_2: D_{A_1A_2} = 2pqd$$

$$A_2A_2: A_{A_2A_2} = \alpha_{A_2} + \alpha_{A_2} = -2p\alpha$$

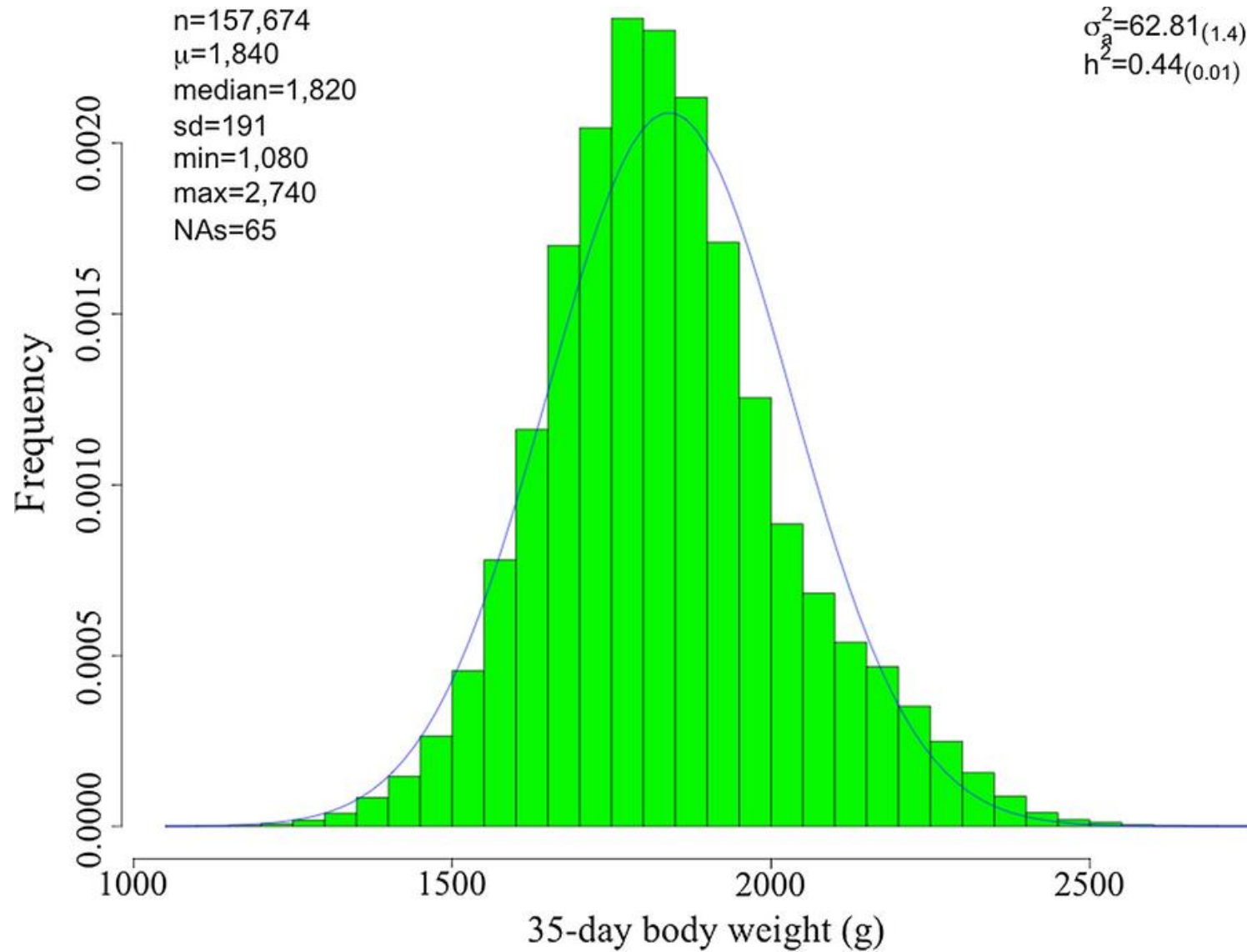
$$A_2A_2: D_{A_2A_2} = -2p^2d$$

AlphaSimR uses dominance degree δ :

$$d = \delta|a|$$

Drawing coefficients from normal distributions (by default):

$$a \sim N(0, 1)$$
$$\delta \sim N(\mu_{DD}, \sigma_{DD}^2)$$



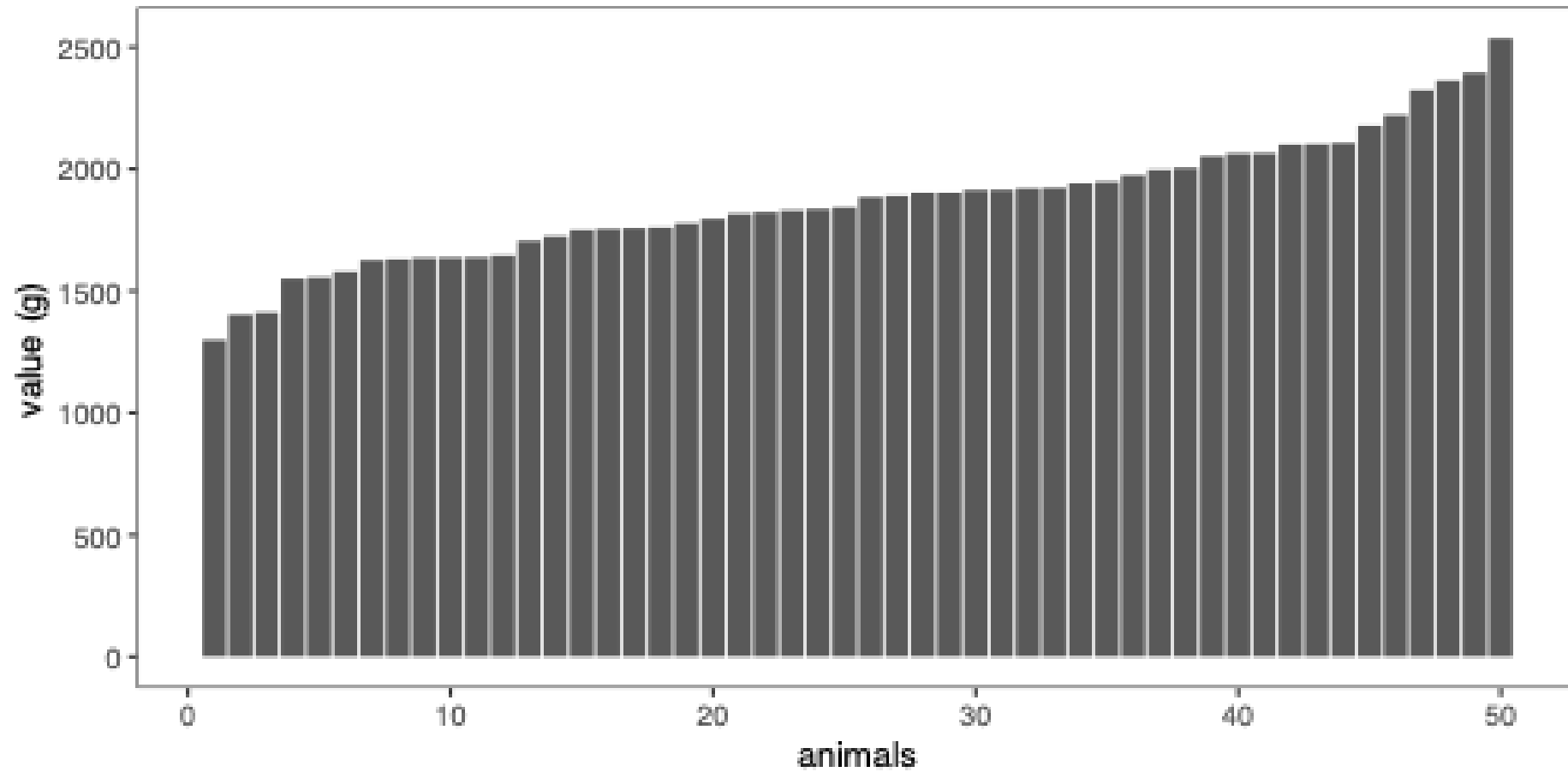
Weight at 35 days in a broiler chicken breeding program (Dadousis et al. 2021).

```
founders <- runMacs(nInd = 1000,  
                   nChr = 10)  
  
simparam <- SimParam$new(founders)  
  
simparam$addTraitAD(nQtlPerChr = 1000,  
                   mean = 1840,  
                   var = 191^2,  
                   meanDD = 0,  
                   varDD = 0.1)  
simparam$setVarE(h2 = 0.44)
```

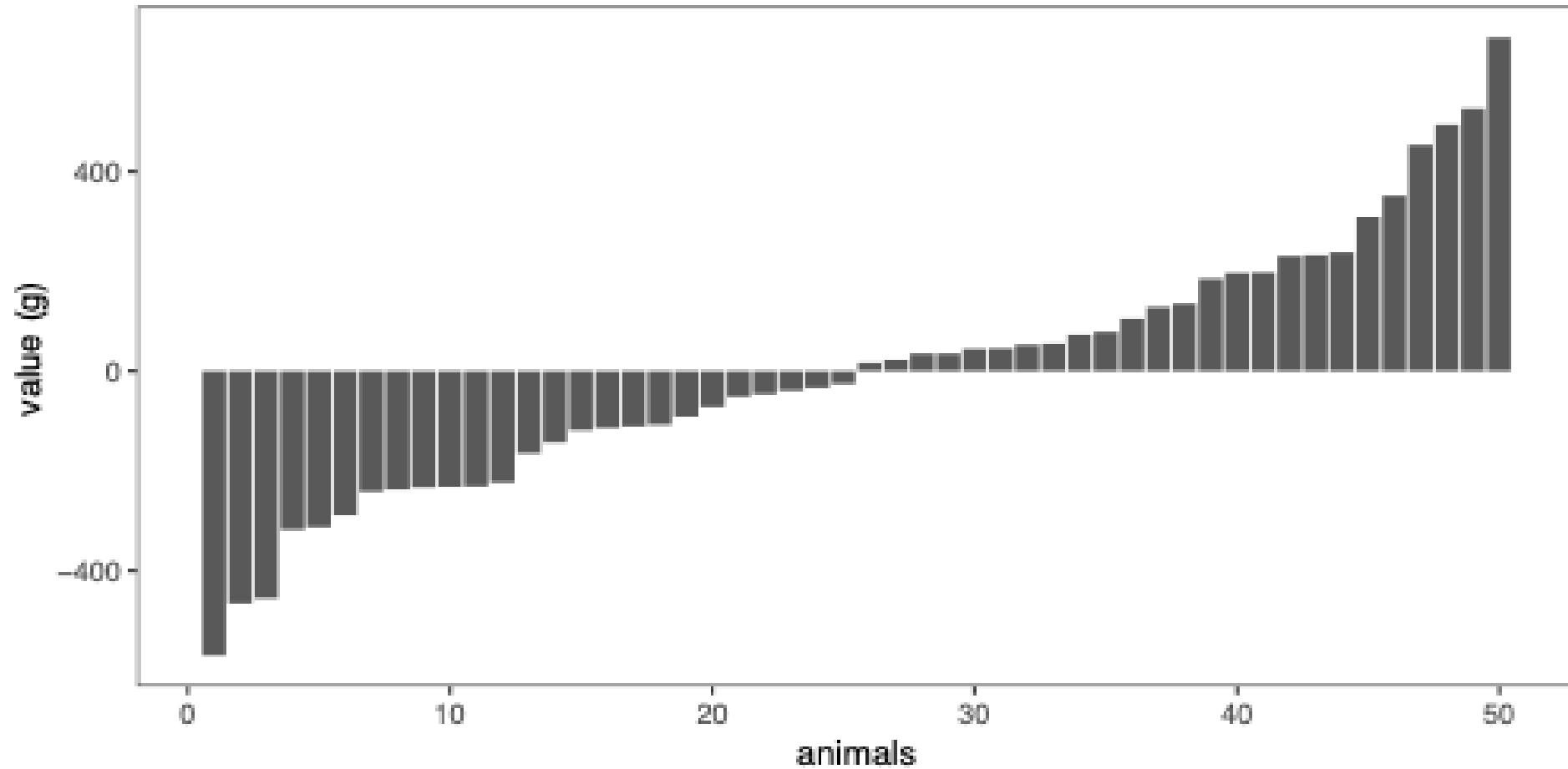
```
pop_full <- newPop(founders,  
                    simParam = simparam)
```

```
pop <- pop_full[1:50]
```

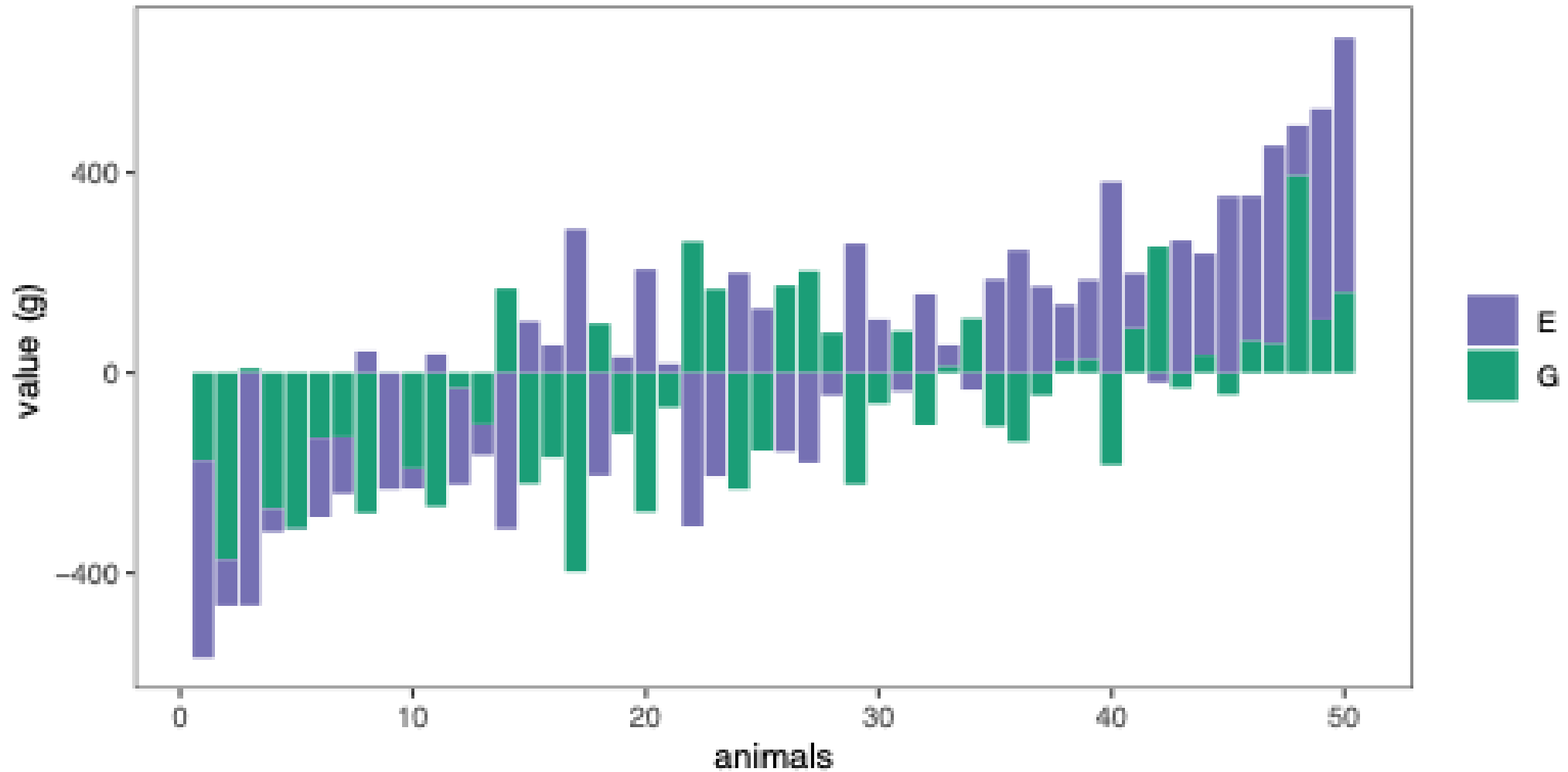
```
stats <- tibble(id = pop@id,  
                pheno = pop@pheno[,1],  
                P = pheno - meanP(pop),  
                gv = gv(pop)[,1],  
                G = gv - meanP(pop),  
                E = P - G,  
                A = bv(pop, simParam = simparam)[,1],  
                D = dd(pop, simParam = simparam)[,1])
```



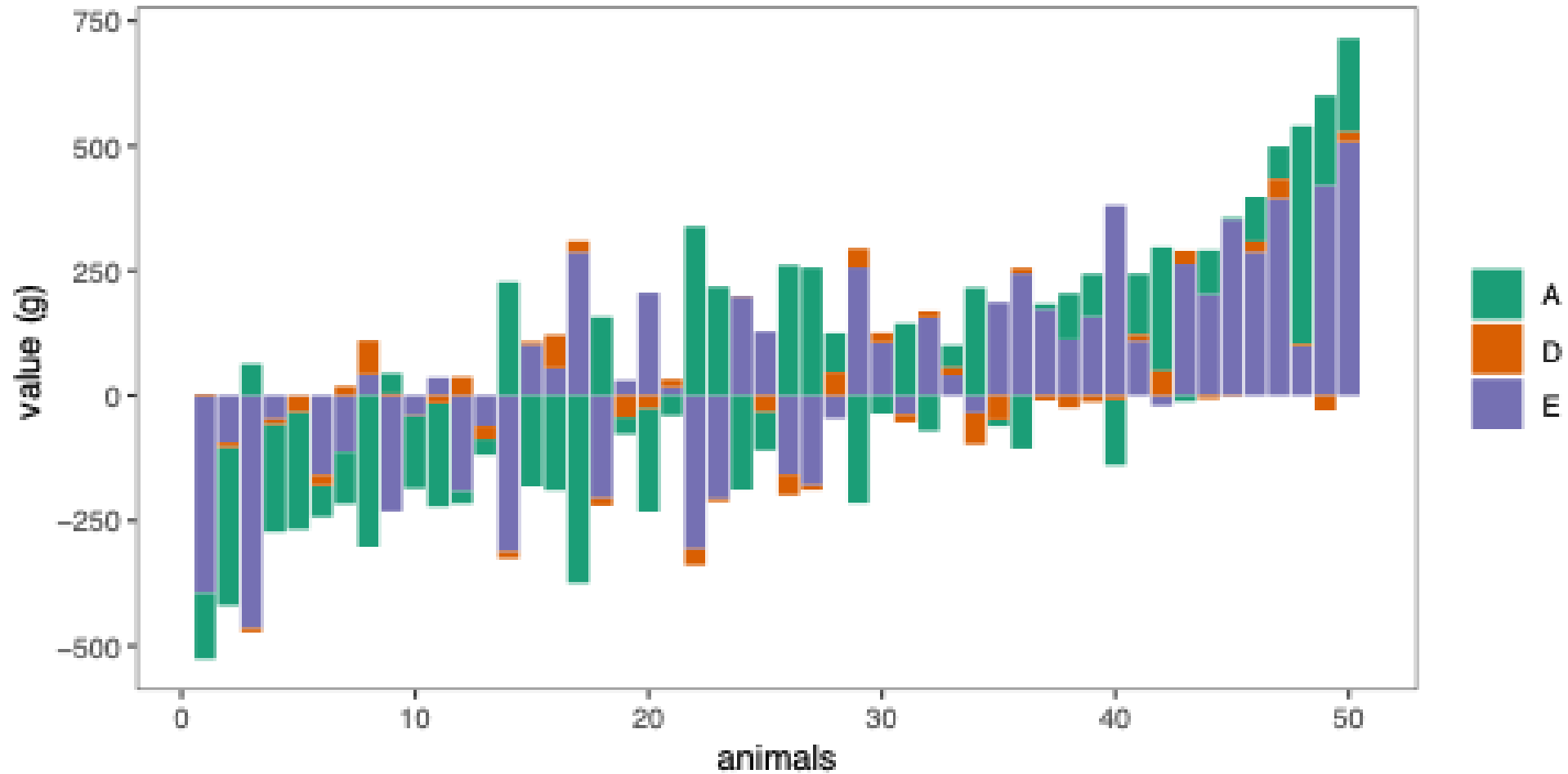
Simulated phenotypic values (sample of 50 animals)



Phenotypic values corrected for population mean
(as they should be in quantitative genetics)



Genetic and environmental values



Additive, dominance and environmental values

```
selected <- selectInd(pop = pop_full,  
                      nInd = 100,  
                      simParam = simparam)  
  
offspring <- randCross(pop = selected,  
                      nCrosses = 1000,  
                      simParam = simparam)
```

Breeders equation (simplest form) predicts the response to one generation of selection and mating.

$$R = h^2 S$$

Where S is the selection differential.

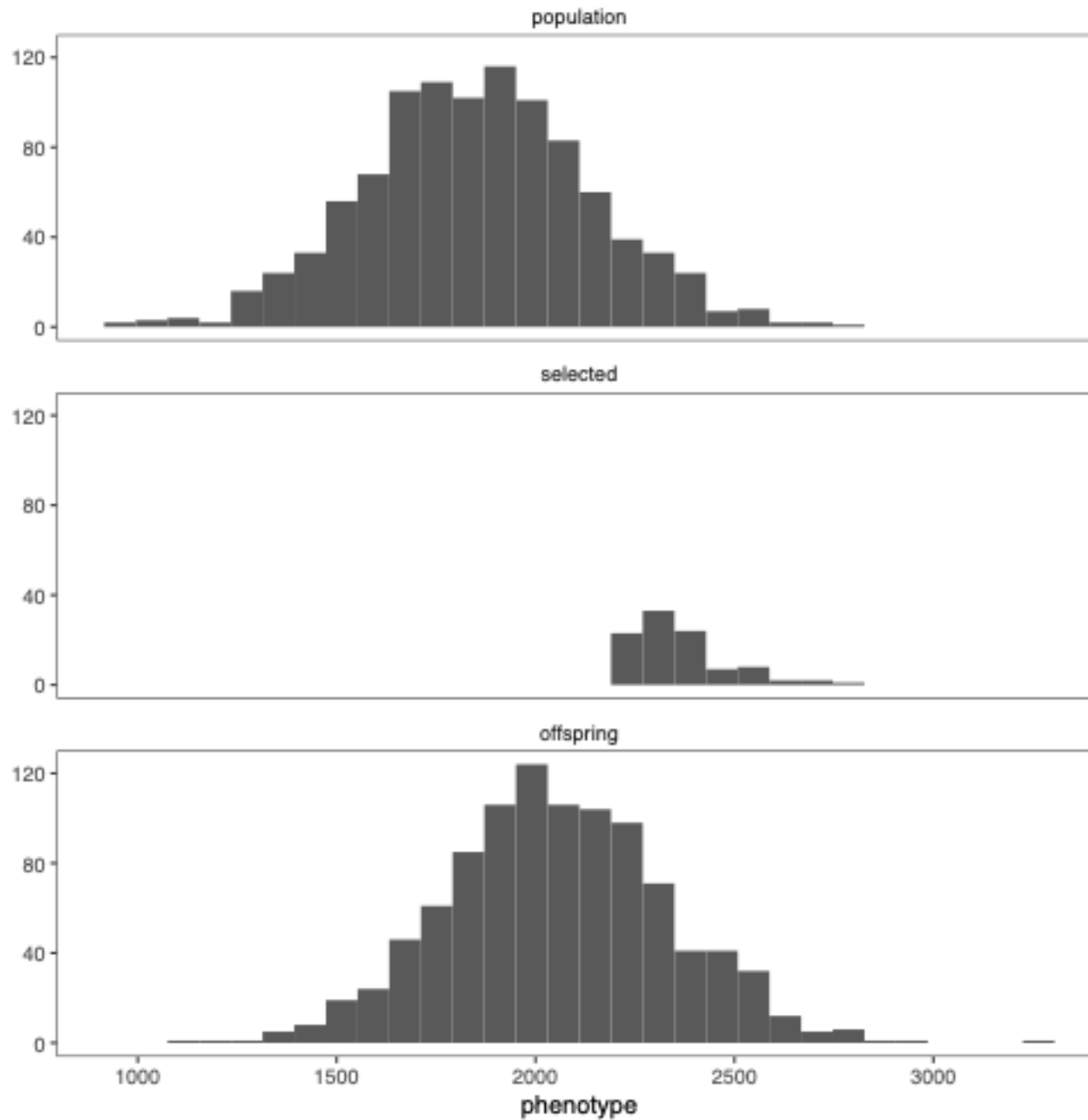
```
mean_selected <- meanP(selected)
mean_all <- meanP(pop_full)

selection_diff <- mean_selected - mean_all

pred_response <- selection_diff * 0.44

mean_offspring <- meanP(offspring)

selection_response <- mean_offspring - mean_all
```



One replicate:

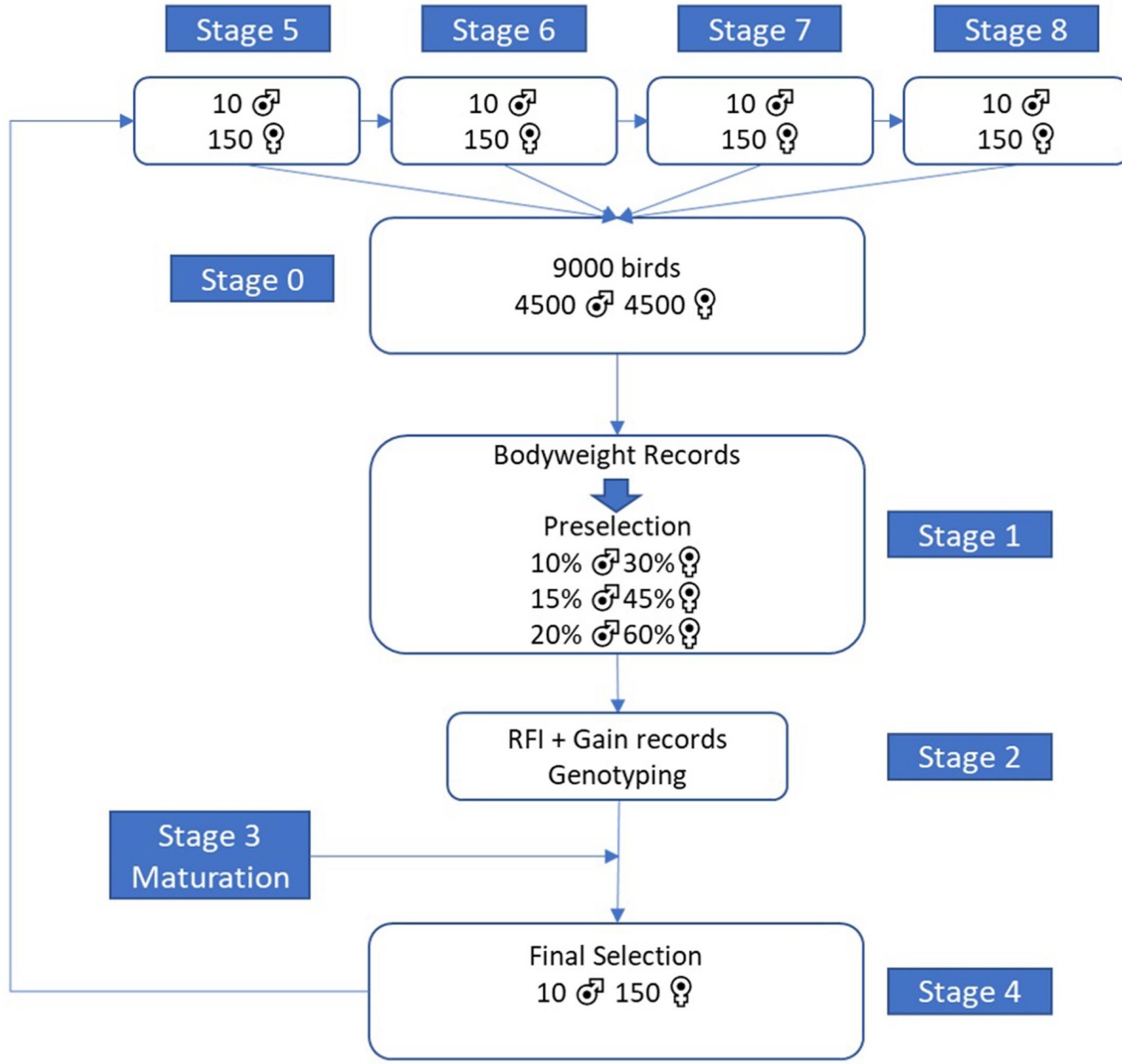
Predicted response: 221 g

Observed response: 240 g

Complex breeding structures: de Hollander et al. (2024)

de Hollander, C.A., Chu, T.T., Marois, D., Felipe, V.B., Lopes, F.B. and Calus, M.P.L. (2025), The Effect of Preselection on the Level of Bias and Accuracy in a Broiler Breeder Population, a Simulation Study. J Anim Breed Genet, 142: 392-407.
<https://doi.org/10.1111/jbg.12908>

”Many breeding programmes have to perform preselection, as genotyping and phenotyping all potential breeder candidates is often not a feasible option. There is need to understand how preselection affects the quality of the genomic estimated breeding values (EBVs) at final selection and thereby can affect genetic progress.”



We will implement the breeding structure, setting aside evaluation for now.

Goal:

- Illustrate complex breeding structures in AlphaSimR
- Prepare for evaluation

```
library(AlphaSimR)
library(ggplot2)
library(patchwork)
library(tidyr)

founders <- runMacs(nInd = 1200,
                   nChr = 10)

simparam <- SimParam$new(founders)

simparam$setSexes("yes_sys")
```

```
trait_correlation_matrix <- matrix(c(1, 0.11, 0.15,  
                                     0.11, 1, 0.14,  
                                     0.15, 0.14, 1),  
                                   ncol = 3,  
                                   byrow = TRUE)  
  
simparam$addTraitA(nQtlPerChr = 100,  
                  mean = c(0, 0, 0),  
                  var = c(1, 1, 1),  
                  corA = trait_correlation_matrix)  
  
simparam$setVarE(h2 = c(0.18, 0.19, 0.17))
```

```
founder_population <- newPop(founders,  
                              simParam = simparam)  
  
males <- selectInd(founder_population,  
                   nInd = 10 * 4,  
                   sex = "M",  
                   use = "rand",  
                   simParam = simparam)  
  
females <- selectInd(founder_population,  
                    nInd = 150 * 4,  
                    sex = "F",  
                    use = "rand",  
                    simParam = simparam)
```

```
sires <- list(males[1:10],  
             males[11:20],  
             males[21:30],  
             males[31:40])  
  
dams <- list(females[1:150],  
            females[151:300],  
            females[301:450],  
            females[451:600])
```

```
## "Stage 0": offspring created from the 4 breeder groups
chicks <- vector(length = 4, mode = "list")

for (group in 1:4) {
  chicks[[group]] <- randCross2(males = sires[[group]],
                                females = dams[[group]],
                                nCrosses = dams[[group]]@nInd,
                                nProgeny = n_progeny_group[group],
                                simParam = simparam)
}

offspring <- mergePops(chicks)
```

```
## "Stage 1": preselection based on body weight (trait 1)
```

```
males_preselected <- selectInd(offspring,  
                                sex = "M",  
                                nInd = 4500 * 0.1,  
                                trait = 1,  
                                use = "pheno",  
                                simParam = simparam)
```

```
females_preselected <- selectInd(offspring,  
                                  sex = "F",  
                                  nInd = 4500 * 0.3,  
                                  trait = 1,  
                                  use = "pheno",  
                                  simParam = simparam)
```

```
accuracy_preselection[step] <- cor(offspring@gv[,1], offspring@pheno[,1])
```

```
## "Stage 4": selection based on index (of true breeding values)
males_selected <- selectInd(males_preselected,
                             nInd = 10,
                             trait = selIndex,
                             b = c(1, 0.82, 0.5),
                             use = "gv",
                             simParam = simparam)

females_selected <- selectInd(females_preselected,
                              nInd = 150,
                              trait = selIndex,
                              b = c(1, 0.82, 0.5),
                              use = "gv",
                              simParam = simparam)
```

```
## Advance stages 5-8  
sires <- list(males_selected, sires[[1]], sires[[2]], sires[[3]])  
dams <- list(females_selected, dams[[1]], dams[[2]], dams[[3]])
```

Evaluation and genomics

Selection in AlphaSimR

- Selection functions `selectInd`, `selectFam`, `selectWithinFam`, `selectCross`
- Argument `trait=` indicates trait number or a name of a function (e.g., index selection with `selIndex`)
- Argument `use=` can be genetic values "gv", breeding values "bv", phenotype "pheno", random "rand" or EBV slot "ebv"
- Population object has `@ebv` slot to store EBVs
- Alternatively: pick individuals manually with hard bracket indexing, e.g. `pop[1:10]`

SNP chips in AlphaSimR

```
simparam$addSnpChip(nSnpPerChr = 100)
```

Selects SNPs from the segregating sites.

By default, they do not overlap causative variants.

Multiple chips possible

Genomic prediction in AlphaSimR

RRBLUP (and RRBLUP2) function implements a ridge regression GBLUP.
setEBV stores predictions in @ebv slot.

```
model <- RRBLUP(training_pop, simParam = simparam)  
evaluated <- setEBV(candidates, model, simParam = simparam)
```

That is usually not enough. External program needed.

Genetic evaluation

Instead of implementing genetic analysis methods, AlphaSimR requires you run your own, if you want realistic evaluation.

Blupf90, optimal contributions, etc.

Data to export:

`pullSnpChipGeno`

`population@pheno`

`simparam$ped`

Image credits

Fernyhough, M., Nicol, C.J., van de Braak, T. *et al.* The Ethics of Laying Hen Genetics. *J Agric Environ Ethics* **33**, 15–36 (2020). <https://doi.org/10.1007/s10806-019-09810-2> <http://creativecommons.org/licenses/by/4.0/>

Zuidhof M, Schneider B, Carney V, Korver D, Robinson F. Growth, efficiency, and yield of commercial broilers from 1957, 1978, and 2005. *Poultry Science*. 2014;93: 2970–2982. <https://doi.org/10.3382/ps.2014-04291>
<http://creativecommons.org/licenses/by-nc-nd/4.0/>

Dadousis, C., Somavilla, A., Ilska, J.J. *et al.* A genome-wide association analysis for body weight at 35 days measured on 137,343 broiler chickens. *Genet Sel Evol* **53**, 70 (2021). <https://doi.org/10.1186/s12711-021-00663-w>
<http://creativecommons.org/licenses/by/4.0/>

de Hollander, C.A., Chu, T.T., Marois, D., Felipe, V.B., Lopes, F.B. and Calus, M.P.L. (2025), The Effect of Preselection on the Level of Bias and Accuracy in a Broiler Breeder Population, a Simulation Study. *J Anim Breed Genet*, 142: 392-407. <https://doi.org/10.1111/jbg.12908> <http://creativecommons.org/licenses/by/4.0/>



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